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Personalized Federated Learning for Equitable Breast Cancer Detection in Underrepresented Pacific Islander Populations

Lydia J. Sollis[1,2] · Patricia M. Young[3] · Arianna Bunnell[1,2] · Brandon Quon[1] · Brenda Y. Hernandez[1] · Thomas K. Wolfgruber[1] · John A. Shepherd[1]

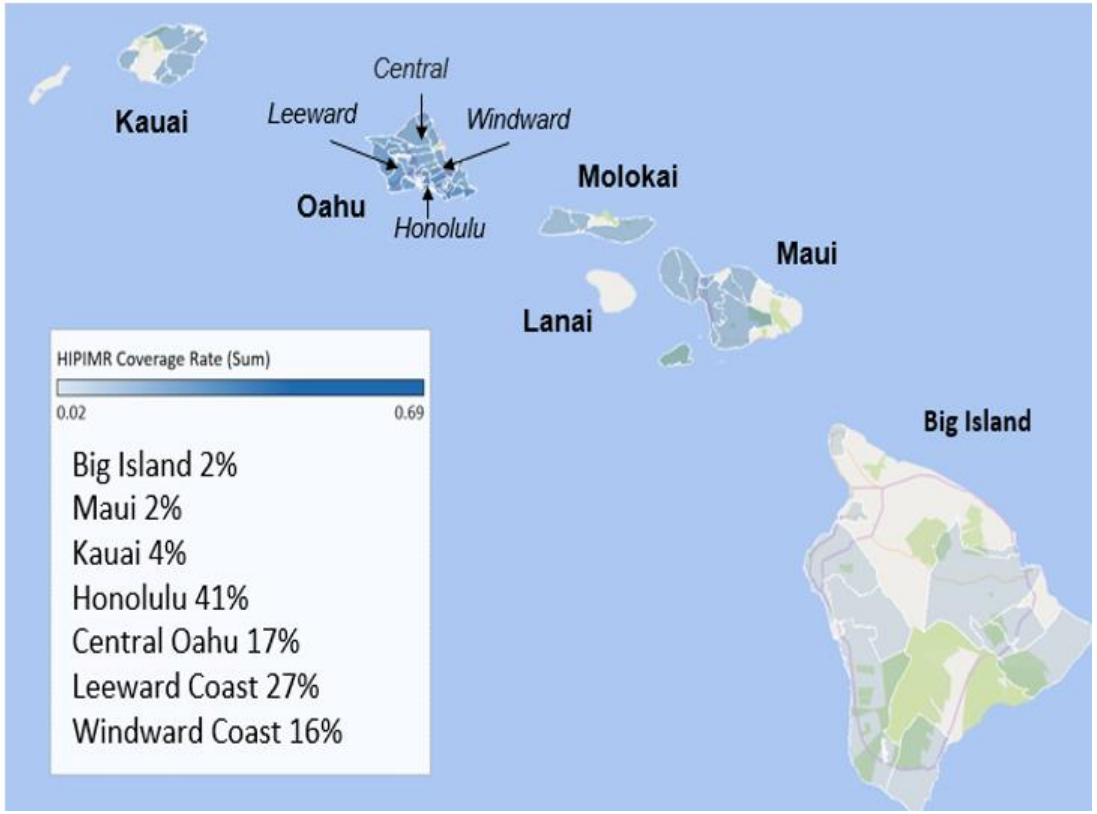
University of Hawai'i Cancer Center · University of Hawai'i at Mānoa · Hawai'i Pacific University

lsollis@cc.hawaii.edu

Why Federated Learning Matters for Hawai‘i

The University of Hawai‘i Cancer Center is the only NCI-designated cancer center in the Pacific, serving a geographically isolated and diverse population across Hawai‘i and the wider Pacific.

Federated learning could allow Pacific sites and larger external cancer centers to train models without pooling patient data, while respecting Indigenous data sovereignty.



HIPIMR imaging coverage across the Hawaiian Islands, by home-address region (2009–present).

HIPIMR at a Glance

Registry: prospective cohort of >125,000 women imaged 2009–present, linked annually to the Hawai‘i Tumor Registry.

Imaging volume: 568,429 total visits — 354,028 mammograms, 116,675 tomosynthesis, 77,352 ultrasound, 6,158 MRI.

Confirmed cancers: 6,571 invasive (8,375 any), via the same registry linkage.

https://hipimr.shepherdresearchlab.org/ · R01CA263491

[11] Loo LWM, Williams M, Hernandez BY. The high and heterogeneous burden of breast cancer in Hawaii: a unique multiethnic U.S. population. Cancer Epidemiol. 2019;58:71–76.

The Problem: Federation Does Not Guarantee Equal Benefit

- Federated learning in mammography has focused largely on breast-density classification; cancer-detection studies remain limited, and prior work has not evaluated personalized FL in Pacific Islander populations [20–23].

Research Question

Does personalized FL improve worst-site performance and reduce cross-site variation compared with shared-global FL in a heterogeneous mammography federation?

Federated learning (FL) enables multi-site training without sharing patient data,

but standard FedAvg [13] produces one global model that can underperform at sites that differ in population, case mix, and acquisition.

Personalized FL instead lets each site keep a model adapted to its own distribution while still benefiting from shared learning.

[13] McMahan B, et al. Communication-efficient learning of deep networks from decentralized data. AISTATS 2017:1273–1282.

A Heterogeneous Three-Site Federation

	Site A HIPIMR (Hawai‘i)	Site B RSNA-US	Site C RSNA-AU
Exams (cancer)	2,188 (520)	996 (249)	936 (234)
NHPI representation	16.2%	0.8%	n/a
Scanner(s)	Hologic (~96%)	7 units (Hologic)	3 units
Hardware	V100 (DGX-1)	GCP VM, T4	RTX A4000

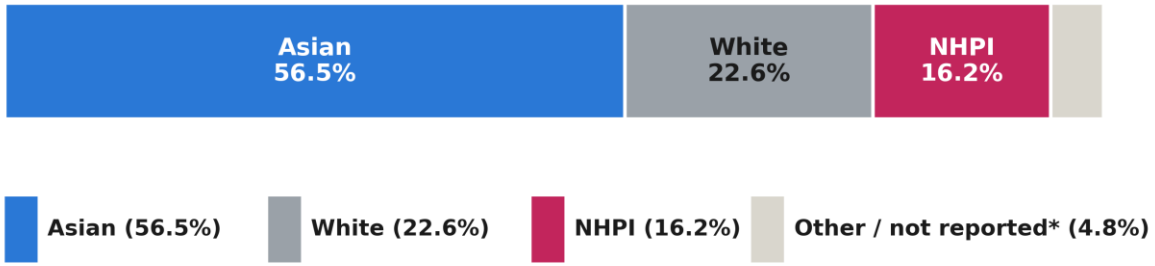


Fig. 1. Site A (HIPIMR) test-set composition by self-reported race/ethnicity, n=878 breasts (this study).
**Includes small-n groups pooled for privacy.*

Labeling Across Sites

- Site A (HIPIMR): controls matched 3:1 to registry-confirmed invasive + in-situ cancers on age, density, and scanner — Hawai‘i population.
- Sites B/C (public RSNA dataset [24]): screen-detected, biopsy-confirmed cancers, trimmed to the same 3:1 control-to-case ratio — mainland U.S. (Site B) and Australian (Site C) population.

[24] Trivedi HM, et al. Open-source dataset for the RSNA Screening Mammography Cancer Detection Challenge. Radiol Artif Intell. 2026:e250375.

FL Strategies Compared

Centralized *	Single model on pooled 3-site data (pooled-data benchmark).
Local-only *	Each site trains GMIC independently with only site-local data
FedAvg [13]	Dataset-size-weighted averaging into one shared global model.
FedProx [14]	FedAvg + proximal penalty (μ=0.1) limiting client drift.
FedBN [15]	FedAvg with batch-normalization kept local per site.
Ditto [16]	Global model + regularized per-site personalized model vk.
Module-wise Ditto	Ditto with a separate λ per GMIC module (global / local / fusion).

** Local (non-federated) training strategies*

[13] McMahan et al., AISTATS 2017 · [14] Li T et al., MLSys 2020 · [15] Li X et al., arXiv:2102.07623 (2021) · [16] Li T et al., ICML 2021

How Ditto Personalizes

Ditto [16] jointly trains a shared global model w* (via FedAvg [13]) and a personalized model vk per site:

minimize over vk: $F_k(v_k) + (\lambda/2)\cdot\|v_k - w^*\|^2$

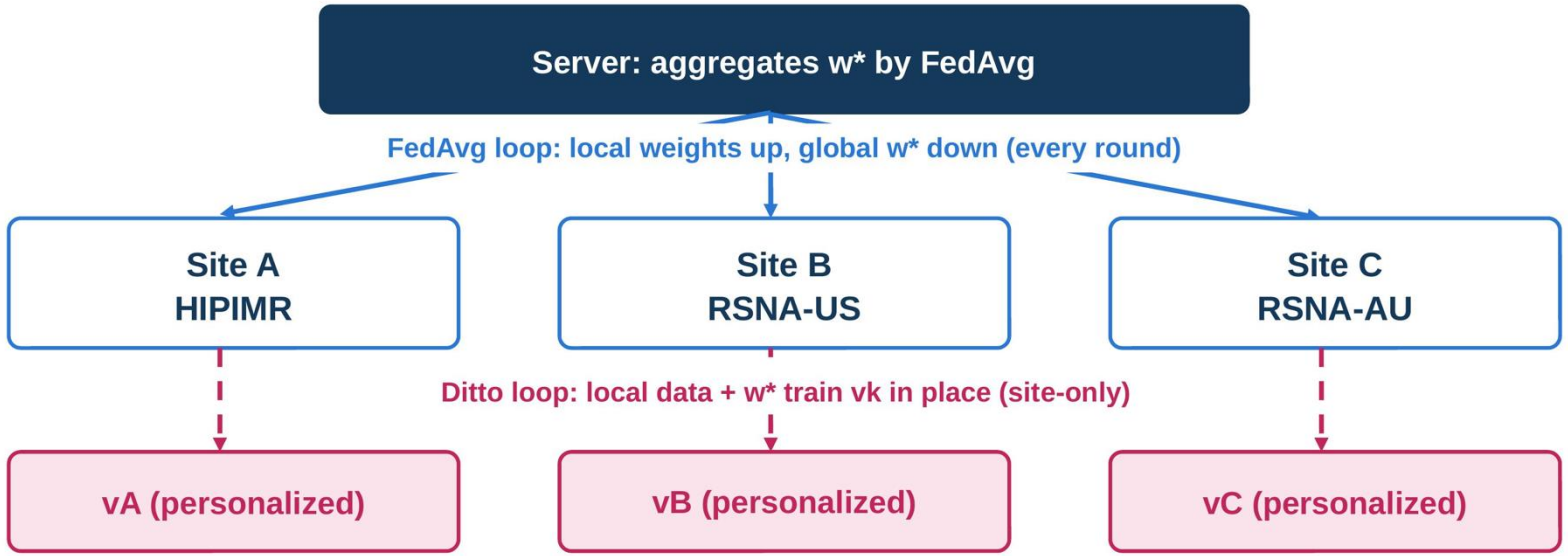


Fig. 2. Ditto's two training loops. Blue: the shared FedAvg loop through the server, each round. Magenta: a loop that runs only at each site, fine-tuning a personalized v_k toward w on that site's own data; v_k is never sent to the server.*

Larger λ pulls a site's model back toward the shared global model; smaller λ allows more local adaptation. A single λ, swept over {0.01, 0.05, 0.1, 0.5, 1, 2} on validation data and chosen to maximize worst-site AUC, is applied at every site; each site deploys its personalized vk, not the global model.

[16] Li T, Hu S, Beirami A, Smith V. Ditto: Fair and robust federated learning through personalization. ICML 2021:6357–6368.

GMIC Architecture & Fine-Tuning

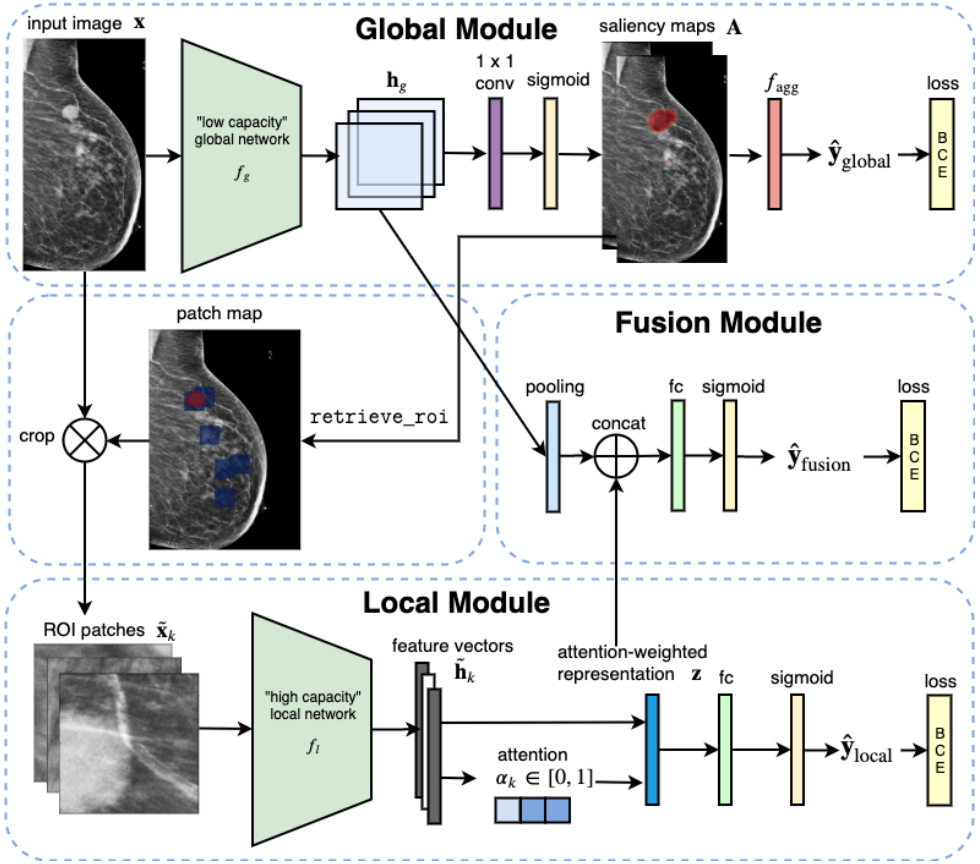


Fig. 3. GMIC's three-module architecture at 2944 × 1920 resolution: a global module scans the whole image for saliency, a local module examines the K=6 highest-attention ROI patches at full capacity, and a fusion module combines both. Adapted from [2,3].

Fine-Tuning (based on published pretrained weights)

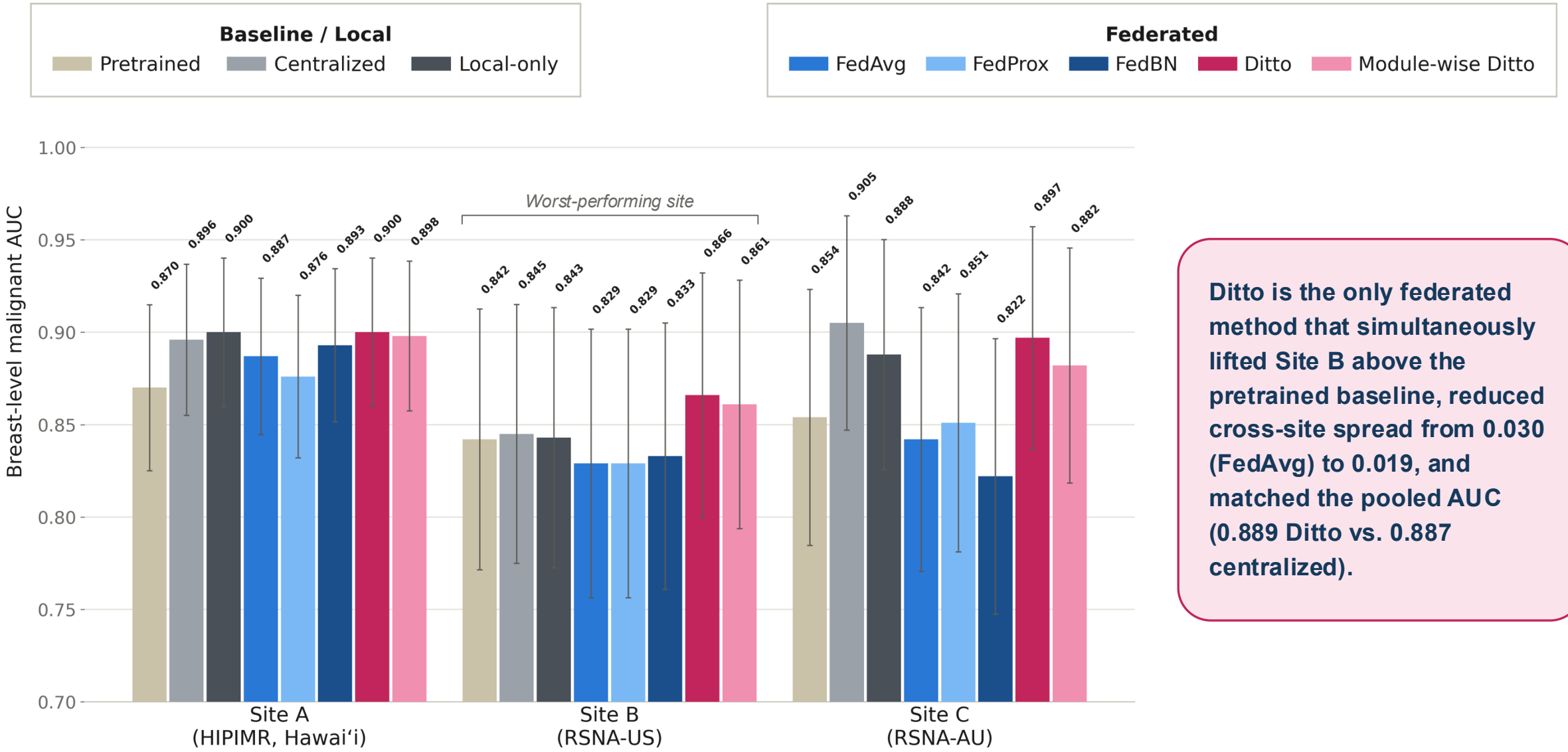
- Train/validation/test split 70/10/20%; class imbalance (7.2:1 view-level pos:neg) handled via class-balanced sampler, positive-class weight 1.0
- Adam: lr 3×10^{-5} (heads) / 3×10^{-7} (backbone), weight decay 1×10^{-6}, 2-epoch initial backbone freeze.
- Local models: up to 40 epochs, batch size 32, early stopping on validation AUC (patience 6).
- Federated: 60 rounds × 1 local epoch/round from the same pretrained init; FedProx μ=0.1, Ditto λ = 0.05, module-wise Ditto λ (global/local/fusion) = 0.01, 0.5, 0.0; NVFLARE 2.5 used for federated training

Evaluation & Cross-Site Performance Variation

- Breast-level AUC, averaging CC + MLO view predictions per breast, with DeLong 95% confidence intervals [26].
- Deployed round chosen by validation AUC: pooled for shared-global models, site-specific for personalized ones.
- Cross-site AUC spread = sample SD of site-level AUC; lower values indicate more uniform site performance (not a demographic-equity measure by itself).
- Site A subgroup analysis at fixed operating points: sensitivity at 90% specificity, specificity at 90% sensitivity.

[2] Shen Y, et al. An interpretable classifier for high-resolution breast cancer screening images utilizing weakly supervised localization. Med Image Anal. 2021;68:101908.

Personalization Improved Worst-Site AUC



Ditto is the only federated method that simultaneously lifted Site B above the pretrained baseline, reduced cross-site spread from 0.030 (FedAvg) to 0.019, and matched the pooled AUC (0.889 Ditto vs. 0.887 centralized).

[26] DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated ROC curves: a nonparametric approach. Biometrics. 1988;44(3):837–845.

Exploratory Site A (HIPIMR) Subgroup Analysis

Group	Pretrained AUC	Sn@Sp90	Sp@Sn90	Ditto AUC	Sn@Sp90	Sp@Sn90
Overall (878, 104)	0.870 (0.825–0.915)	0.683	0.532	0.899 (0.859–0.939)	0.760	0.672
NHPI (142, 20)†	**0.839 (0.727–0.951)**	**0.600**	**0.393**	**0.898 (0.805–0.991)**	**0.800**	**0.787**
Asian (496, 66)	0.882 (0.828–0.936)	0.667	0.605	0.900 (0.849–0.951)	0.758	0.593
White (198, 11)†	0.825 (0.672–0.978)	0.727	0.492	0.870 (0.733–1.000)	0.818	0.690

n = (group size, positive breasts). AUC columns show 95% CIs (Hanley-McNeil). NHPI (highlighted) pretrained AUC was 0.839 and increased to 0.898 after fine-tuning. †small-n groups are exploratory.

At fixed 90% specificity, Ditto raised NHPI sensitivity 0.60 to 0.80; at fixed 90% sensitivity it raised NHPI specificity 0.39 to 0.79, the largest operating-point gain of any group.

NHPI performance improved after fine-tuning, but similar gains occurred across federated strategies; this analysis does not isolate a personalization effect.

[19] Nguyen DL, et al. Patient characteristics impact performance of AI algorithm in interpreting negative screening DBT studies. Radiology. 2024;311(2):e232286.

Takeaways & Limitations

Key Takeaways

- Federated methods did not consistently outperform local fine-tuning
- Ditto produced the strongest worst-site and overall federated performance, matching centralized non-federated training in point estimates
- Module-wise Ditto did not improve standard Ditto; Site A subgroup gains were not specific to personalization

Limitations

- Single training runs, fine-tuning vs training from scratch, and modest site counts produced wide uncertainty
- 1 epoch per round may have limited algorithm variability
- Individual race/ethnicity data available only at the largest, best-served site

Future Work

- Rerun with:
  - Harmonized demographic labels at all sites
  - HIPIMR as smallest site
  - Unique λ per dataset
  - Additional model architectures/types
  - More epochs per round
- Explore more advanced personalization methods

Personalized FL may be particularly useful when heterogeneous sites do not benefit equally from one shared global model.

[16] Li T, Hu S, Beirami A, Smith V. Ditto: Fair and robust federated learning through personalization. ICML 2021:6357–6368.

Thank you · Questions welcome

Lydia J. Sollis · lsollis@cc.hawaii.edu · Shepherd Research Lab, University of Hawai‘i Cancer Center

Code: github.com/shepherd-lab/GMIC-FL



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